



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE
WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MAC 206: ACTUARIAL MATHEMATICS I

Date: 6th July, 2018

Time: 8.30 – 11.30 am

INSTRUCTIONS

- Answer question ONE and any other TWO questions.



Question 1 (30 marks)

- (a) $\mu(x) = 0.01$ evaluate ${}_{2|2}q_{20}$ (5 marks)
- (b) Show that the force of mortality for De Moivre law is $1/(\omega - x)$ (5 marks)
- (c) Assuming De Moivre's survival function with $\omega = 100$ and $\delta = 0.1$, calculate $\bar{A}_{30:10}^1$ (8 marks)
- (d) Assume $\mu(x) = \mu$ and force of interest of δ , Find $\bar{A}_x$ (5 marks)
- (e) Let $A_x = u, A_{x+n} = s, A_{x:n} = w$. Find $A_{x:n}^1$ (7 marks)

Question 2 (20 marks)

- (a) If $q_{70} = 0.04$ and $q_{71} = 0.05$. Calculate the probability that (70) will die between ages 70.5 and 71.5 under the assumption that death are uniformly distributed within each year of age. (10 marks)
- (f) For a country for a typical person of age 50, the probability ${}_{30}P_{50} = 0.5$
- (i) Find the same probability for a country where the force of mortality for people of age 50 and older is three times higher.
- (ii) Find the same probability for a country where the force of mortality for people of age 50 and older is 0.01 less. (10 marks)

Question 3 (20 marks)

- (a) $\delta = 3\%$ $A_{30} = 0.2, A_{65} = 0.44$ while for $\delta = 6\%$ $A_{30} = 0.09, A_{65} = 0.4235$
For $\delta = 3\%$, compute the APV of annuities of 100,000 paid at the beginning of each year to a person starting from his 30 years, until death. Find the variance of the present value. (10 marks)
- (b) If ${}_{35}p_{30} = 0.9$ find $A_{30:35}$ using information in (a) (10 marks)

Question 4 (20 marks)

- (a) Let X be exponential and $E(X) = 75$. Find ${}_{20|10}q_{50}$ (10 marks)
- (b) For a certain population ${}_{35}p_{30} = 0.95$ and for the rate $\delta = 0.03$, $\bar{A}_{30} = 0.2$ and $\bar{A}_{65} = 0.44$ find $A_{30:35}^1, A_{30,35}$ (10 marks)

Question 5 (20 marks)

- (a) If the force of mortality $\mu_x(t), 0 \leq t \leq 1$ changes to $\mu_x(t) - c$ where c is a positive constant, find the value of c for which the probability that (x) will die within a year will be halved. (10 marks)
- (b) Derive recursive formula for $E(T(x))$ (10 marks)



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MAIN CAMPUS

MAC 210: INVESTMENT AND ASSET MANAGEMENT

Date: 2nd July, 2018

Time: 3.30- 6.30 pm

INSTRUCTIONS

- Answer question ONE and any other TWO question.



Question One (30 Marks)

- a) Give a brief explanation of the following concepts
- Antiques
 - Zero-coupon bonds
 - Indenture
 - Certificate of deposit
 - Commercial papers [5 Marks]
- b) Give and explain five factors an investor keeps in view while deciding about making investment in any particular field [5 Marks]
- c) A company issues a 10-year bond with a coupon rate of 10% paid semi-annually. The bond is callable at par after 5 years. Find the price that guarantees a return of 12% compounded semi-annually for the investor [5 Marks]
- d)
- Give and explain five basic features of the bonds [5 Marks]
 - The return on securities A and B are given below:-

Probability	Security A	Security B
0.5	4	0
0.4	2	3
0.1	0	3

Give the security of your preference. The security has to be selected on the basis of return and risk [5 Marks]
- e)
- Give and explain four methods the investor may use to minimize risk exposure [4 Marks]
 - Explain the concept sweat equity [1 Mark]

QUESTION TWO (20 MARKS)

- a)
- Give and briefly explain four types of debentures [4 Marks]
 - Determine the price of \$1000 zero-coupon bond with yield to maturity of 18% and 10 years to maturity. What is year-to-maturity of this bond, if its price is \$220. [4 Marks]
- b)
- A riskless zero-coupon bond with 10 years-to-maturity has a face value of \$500 and sells for \$ 192.77. Find:-
 - The nominal yield on the bond
 - The yield on the bond if the inflation rate over the life of the bond is 5%
 - The yield if the inflation rate is 8%. [6 Marks]
 - Give and explain the three types of warrants. [3 Marks]
 - Give and explain the parties involved in the issue of the share in the stock exchange market [3 Marks]



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- [1 Mark]

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- [6 Marks]
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- [3 Marks]
- Give and explain the parties involved in the issue of the share in the stock exchange market
- [3 Marks]

QUESTION THREE (20 MARKS)

- a)
- i. Give and explain five main stages of investment process [5 Marks]
 - ii. You currently have total wealth of \$50000. You may invest your wealth in commercial papers at 6% or invest in a risky prospect that will double your money or halve if with equal probability.
 - 1) What is your expected wealth if you take the risky investment [2 Marks]
 - 2) What is your expected wealth if you take the risk-free investment [2 Marks]
 - 3) What is the risk premium of the risky investment in terms of wealth [2 Marks]
 - 4) What is the variance of wealth of the risk-free asset [2 Marks]
- b)
- i. Explain the concept of sinking fund [1 Mark]
 - ii. Briefly compare the important characteristics of the treasury bill and commercial papers that explain why commercial paper provides higher yield [4 Marks]
 - iii. At an annual rate of compounding of 9% how long does it take for a given sum to become double and triple its original value [2 Marks]

QUESTION FOUR (20 MARKS)

- a)
- i. Give and explain methods the investor may use to minimize risk exposure [4 Marks]
 - ii. The risk-free rate is 6%. The risky asset offers an expected return of 15%. Calculate the expected return on a portfolio of the two assets with the following percentages invested in the risky asset
 - 1) 20%
 - 2) 40%
 - 3) 60%
 - 4) 80%
 - 5) 50%[5 Marks]
 - iii. Explain the concept hedge against inflation [1 Mark]
- b)
- i. Give and briefly explain four sources of internal business risk [5 Marks]
 - ii. Give the major determinants of the level of interest rates [3 Marks]
 - iii. Explain concepts Consol and Risk averse [2 Marks]

QUESTION FIVE (20 MARKS)

- a)
- i. Give and explain features of sound capital structure [5 Marks]

- ii. The capital structure of two companies X and Y is given below:-

	X(\$)	Y(\$)
Ordinary Share Capital	100000	25000
Reserves	150000	25000
Sub-Total	250000	50000
10% Debentures	-	200000
Grand Total	250000	250000

Determine the gearing using two usual formulae

[4 Marks]

- iii. A company has an annual demand for material x of 250 tonnes per annum. The cost price per tonne is \$ 20 and stock holding is 25% per annum of the stock value. Delivery cost per batch is 4 dollars. Calculate the economic batch quantity

[4 Marks]

b)

- i. Give five ratios which may aid investment management in finding the balance between efficiency and liquidity
- ii. Explain the term structures of interest rates

[5 Marks]

[2 Marks]

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